

Building an Underwater Vessel Detector

Applying classical machine learning models for underwater acoustic source detection

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Abstract

Maritime domain awareness (MDA) and continuous environmental monitoring are critical operational requirements for protecting New Zealand’s territorial waters and Exclusive Economic Zone (EEZ). Passive acoustic monitoring (PAM) provides a non-intrusive, scalable method to track vessel traffic, yet automated classification remains severely challenged by non-stationary coastal ambient noise and intense temporal autocorrelation. This investigation addresses these vulnerabilities by establishing a transparent, computationally efficient baseline source-versus-noise detector utilizing classical machine learning models optimized for resource-constrained edge surveillance hardware.

Using unclassified continuous acoustic recordings from three bottom-mounted Soundtrap 300HF hydrophones deployed along the Rangitoto Channel in Auckland’s Hauraki Gulf, raw waveforms are mapped across high-dimensional spectral and temporal manifolds. A three-stage dimensionality filter leveraging Spearman rank correlation clustering and mutual information metrics compresses the acoustic space from 148 raw parameters to 76 optimized predictors spanning Low-Frequency Analysis and Recording (LOFAR), Detection of Envelope Modulation on Noise (DEMON), and noise-normalized broadband fields. To eliminate spatial-temporal distribution leakage and ensure true cross-environmental generalization, a strict chronological day-based validation block split is enforced.

Empirical modeling demonstrates that classical tree-based ensembles achieve high classification fidelity, significantly outperforming deep neural networks on out-of-sample environments while maintaining minimal computational overhead. Hyperparameter-tuned XGBoost and CatBoost architectures achieve peak global validation accuracies of 98.41% and 98.27%, respectively. By integrating continuous spatial-acoustic label confidence weights directly into the optimization loss function, the confidence-weighted XGBoost framework stabilizes decision boundaries against volatile littoral backgrounds, securing a Class 1 (Vessel) Precision of 0.99 and a Class 0 (Noise) Recall of 0.99. Furthermore, non-parametric bootstrap resampling confirms that a standalone, shallow Decision Tree preserves exceptional detection capabilities (Mean Macro $F_1 = 0.9773$, 95% CI: [0.9756, 0.9788]). This baseline can be compiled into simple low-level nested conditional logic, proving that traditional estimators offer a mathematically robust, operationally viable blueprint for low-power, real-time uncooperative maritime surveillance at the embedded edge.

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List of Acronyms and Definitions

Term / Acronym	Definition and Operational Explanation
AIS	Automatic Identification System: A cooperative tracking network where ships broadcast GPS telemetry, identity, and speed via radio transponders to prevent collisions and inform port authorities.
Bathymetry	The mapping and measurement of underwater depth and topography (the landscape of the ocean floor), which heavily affects how sound waves bounce and travel.
Broadband Noise	Acoustic energy spread across a wide, continuous range of frequencies. In shipping, this resembles a rushing hiss and is caused by friction between the water and the hull or propeller spray.
Cavitation	The formation and violent collapse of microscopic vapor bubbles caused by the rapid pressure drops behind spinning propeller blades. This is the primary source of a ship's underwater acoustic signature.
CPA	Closest Point of Approach: The exact moment or geographic coordinate where a moving target reaches its minimum distance to a stationary hydrophone station.
Dark Vessel	A ship navigating with its mandatory AIS transponder deactivated, manipulated, or spoofed to evade detection by civil authorities.
DEMON	Detection of Envelope Modulation on Noise: A signal processing method that filters out high-frequency rushing sounds to isolate the rhythmic, cyclic pulse of a ship's propeller blades and shaft rotation.
EEZ	Exclusive Economic Zone: A maritime boundary extending up to 200 nautical miles from a nation's coast, granting sovereign rights over resource extraction, fishing, and conservation.
Hydrophone	A specialized underwater microphone designed to convert microscopic water pressure changes into electrical signals for acoustic analysis.
IUU	Illegal, Unreported, and Unregulated Fishing: Illicit or black-market fishing operations that threaten marine ecosystems, often conducted by dark vessels.
Littoral Zone	Shallow, coastal marine environments close to land. These zones are acoustically complex due to breaking waves, changing tides, and coastal boat traffic.
LOFAR	Low-Frequency Analysis and Recording: A spectral visualization technique used to identify narrow-band, continuous humming frequencies (tonals) generated by internal rotating machinery, such as pumps or onboard generators.
MDA	Maritime Domain Awareness: The comprehensive tracking and security assessment of all activities, vessels, and environmental threats occurring across a designated body of water.
MFCC	Mel-Frequency Cepstral Coefficients: A mathematical method of compressing audio data into texture codes that mimic how human hearing perceives pitch, used to isolate hull resonance profiles.
Narrowband Tonal	A distinct, continuous whistle or hum concentrated at a single precise frequency, acting as an acoustic fingerprint for an engine component.
PAM	Passive Acoustic Monitoring: A stealthy, non-intrusive surveillance method that relies entirely on listening to the existing sounds in an environment without emitting active sonar pings.
PSD	Power Spectral Density: A graph or mathematical array mapping exactly how much acoustic energy or loudness is distributed across individual pitch bins from bass to treble.
SNR	Signal-to-Noise Ratio: The measurement of how loud a targeted signal (like a vessel's engine) is relative to the volume of the ambient ocean background (like wind, rain, or snapping shrimp).
SSP	Sound Speed Profile: A vertical map showing how the speed of sound shifts at various ocean depths due to changing water temperatures, pressures, and salinity layers.
UATR	Underwater Acoustic Target Recognition: The automated technical discipline of deploying software algorithms to detect, track, and categorize ships using underwater sound recordings.

Chapter 1

Introduction

1.1 Project Motivation

The primary motivation for this research stems from the operational vulnerabilities inherent in relying entirely on cooperative tracking frameworks within maritime domain awareness (MDA) infrastructures. Automated tracking systems provide an independent, verifiable method to observe coastal choke points, monitor compliance in marine reserves, and detect uncooperative targets that avoid radio transponder requirements. By establishing a computationally viable acoustic tracking baseline, this research provides a data-driven path to supplement maritime surveillance architectures without the immediate overhead of deep learning deployments.

1.2 Background

1.2.1 Signal Processing of Acoustic Hydrophones

The ocean environment presents a complex acoustic medium. Unlike radar or optical sensors, which degrade rapidly underwater, acoustic waves propagate over long distances. This characteristic makes hydrophones the primary sensor for underwater surveillance. A typical deployment topology, such as the three Soundtrap 300HF underwater sound recorders deployed along the Rangitoto Channel, continuously records scalar pressure variations.

Ocean ambient noise is inherently non-stationary. Extracting actionable data from this raw input requires consistent signal processing techniques. This process involves isolating specific acoustic signatures, such as the broadband noise generated by propeller cavitation, from background noise. Analysts extract and tabulate features from multi-node hydrophone data. Time-frequency transforms are used to convert one-dimensional signals into representations like spectrograms for downstream binary classification.

1.2.2 Strategic Defence and Marine Protection

For defence organisations like the New Zealand Defence Force (NZDF), operating in a technologically advanced environment requires evidence-based scientific metrics. Securing the Exclusive Economic Zone (EEZ) and maintaining maritime domain awareness are core components of national security pipelines. Relying solely on cooperative tracking systems can introduce operational vulnerabilities.

By establishing independent tracking capabilities through acoustic monitoring, nations can verify maritime activity. This assists in mitigating the risk of conflict escalation and

protecting regional resources. For instance, illegal, unregulated, and unreported (IUU) fishing threatens marine sustainability. Detecting these vessels is difficult because operators sometimes turn off, manipulate, or spoof their Automatic Identification System (AIS) transponders to mask illicit operations [8]. Acoustic surveillance provides an uncooperative, complementary detection method. It relies on the physical noise radiated by a ship's propulsion, which cannot be eliminated while the vessel is in motion.

1.2.3 Civilian and Commercial Maritime Applications

Beyond military and border security applications, underwater acoustic detection has civilian utility. Environmental monitoring is a primary beneficiary, allowing authorities to track maritime traffic impacts on marine ecosystems. Furthermore, while commercial shipping is regulated with class A AIS transceivers, a population of smaller recreational and civilian vessels does not transmit AIS data. Automated acoustic detection can provide port authorities and marine traffic controllers with a real-time picture of local waterways. This data supports collision avoidance, search and rescue operational efficiency, and overall navigational safety.

1.3 Problem Statement

Current maritime surveillance and domain awareness frameworks rely heavily on the Automatic Identification System (AIS) to track vessel movements. However, this reliance creates a vulnerability: AIS transmissions are cooperative and can be intentionally disabled, manipulated, or spoofed by actors engaging in illicit activities [8]. Furthermore, a significant volume of smaller maritime traffic does not utilize AIS. While ship acoustic noise offers an alternative for detection, non-stationary ambient noise renders simple threshold-based detection systems ineffective. Therefore, it is necessary to develop automated, data-driven acoustic classifiers capable of distinguishing ship signatures from complex background noise to ensure uncooperative maritime domain awareness.

1.4 Project Aim and Scope

1.4.1 Aim

The overall aim of this project was to develop and evaluate classical machine learning models capable of automatically detecting, and potentially classifying, underwater acoustic sources to improve maritime domain awareness.

1.4.2 Scope

This project was executed over the duration of one semester and focused specifically on the following parameters:

- **Data Utilisation:** The research exclusively used an unclassified dataset from the Hauraki Gulf underwater noise survey. This comprised continuous acoustic data sampled at 16 kHz across three distinct Soundtrap 300HF hydrophone stations.
- **Data Processing:** The project involved exploring a pre-labelled dataset from the acoustic data window and optimizing it for classical machine learning methods.

- **Model Development:** The core technical work focused on training classical machine learning algorithms, specifically Support Vector Machines, Random Forest, and Gradient Boosting, using Python libraries such as Scikit-Learn.
- **Task Boundaries:** The scope of this research was strictly bounded to binary signal-versus-noise detection to establish a reliable baseline framework. Multi-class vessel classification (e.g., distinguishing specific ship types or hulls) was explicitly designated as out of scope for this stage of the investigation.
- **Deliverables:** The expected outputs included the development of a baseline source-versus-noise detector, an error analysis of the results, the identification of dataset limitations with proposed improvements, and a final dissertation. Deep learning approaches (such as Convolutional Neural Networks) were outside the primary scope of this classical machine learning mandate.

Chapter 2

Related Work

2.1 Paradigms in Underwater Acoustic Target Recognition

The field of Underwater Acoustic Target Recognition (UATR) has undergone a structural transition over recent decades, moving away from hand-crafted feature extraction pipelines toward automated, highly parameterized end-to-end deep learning architectures [1]. Meta-analyses of these historical shifts demonstrate that while localized deep classification frameworks achieve exceptional empirical efficacy within controlled, stationary training distributions, their cross-dataset generalization can diverge drastically when deployed across independent geographic basins or volatile operational environments [1]. Deep neural network classifiers frequently adapt to localized acoustic transfer functions, complex bathymetry, and seasonal sound speed profiles rather than isolating invariant target mechanics, essentially memorizing channel states instead of extracting true source acoustic primitives [1].

To overcome these generalization collapses, a paradigm shift has emerged within modern hydroacoustic machine learning that advocates for the integration of physics-guided domain knowledge directly into the statistical modeling process [12]. Rather than relying on uninterpretable, data-hungry black-box transformers or deep networks, contemporary architectures embed known acoustic propagation mechanics and machinery kinematics straight into the feature extraction layer [12]. This hybrid framework is further augmented by novel temporal modeling dimensions, which project raw one-dimensional hydroacoustic waveforms into high-dimensional geometric spaces to isolate continuous periodic correlations that traditional architectures fail to capture [13]. By constraining the machine learning hypothesis space using structured spectral and temporal manifolds, recent developments demonstrate that classifiers maintain high cross-environmental stability, even when subjected to intense oceanographic fluctuations driven by shifting thermal and salinity boundaries [14].

In sovereign defence and maritime surveillance contexts, this divergence highlights a critical architectural and operational trade-off. Many state-of-the-art deep learning models remain structurally opaque and computationally intensive, requiring high-overhead hardware configurations that complicate real-time, low-latency edge processing on autonomous or battery-powered listening nodes. Furthermore, due to national security sensitivities and the highly classified, proprietary nature of military sonar networks, deep learning applications often function as un-reproducible silos. To complement these opaque frameworks, there is a clear operational requirement for transparent, deterministic baseline systems. Developing robust, tabulated feature spaces using traditional statistical classifiers provides a verifiable performance standard. This ensures that open-source, reproducible machine learning pipelines can be deployed effectively on resource-constrained edge hardware to

maintain continuous, uncooperative maritime domain awareness.

2.2 Deep Learning Deployments and Environmental Baselines in Littoral Channels

The foundational exploration of the Hauraki Gulf dataset was conducted by the Defence Science Technology (DST), establishing the baseline environment and sensor topology utilized in this research [2]. Deployed within Auckland’s Rangitoto Channel, this passive monitoring network captured continuous acoustic signatures across several hundred ship passages using three bottom-mounted Soundtrap 300HF hydrophones [2]. The primary objective of that research was to evaluate the efficacy of deep neural networks specifically comparing a pre-trained Wavegram-Logmel-CNN against a custom-designed convolutional neural network (CNN) to execute binary ship-versus-noise classification [2].

The environmental analysis documented severe non-stationary ambient noise fields endemic to local waterways [2]. High-frequency biological transients, predominantly driven by localized snapping shrimp populations, dominate the ambient spectrum between 1 kHz and 4 kHz with intensities ranging from 70 dB to 80 dB [2]. Additionally, low-frequency tidal flow dynamics and tether strumming below 200 Hz introduce transient acoustic inflation, particularly when large vessels pass the stations [2]. To isolate target signatures from these volatile biological and hydrodynamic backgrounds, the data was downsampled from 288 kHz to 16 kHz, focus-filtering the analysis on the sub-8 kHz band where the majority of ship-radiated machinery noise resides [2].

The empirical performance profiles of these deep architectures revealed a distinct operational asymmetry:

- **The Wavegram-Logmel-CNN Architecture:** Benefiting from structural feature embeddings pre-trained on the expansive AudioSet corpus, this model achieved a Class 1 (Vessel) Precision of 1.00 [2]. However, it exhibited a low Class 1 Recall of 0.263, frequently misclassifying isolated segments of active ship transits as ambient background noise [2].
- **The Custom CNN Configuration:** By sequentially expanding convolutional kernel sizes and filter depths, the custom model achieved an enhanced Class 1 Recall of 0.621 and an overall global validation accuracy of 92.6% [2]. This expansion in network complexity triggered localized overfitting, which required the integration of strict regularization methods, including weight penalties and dropout layers, to maintain stability [2].

While these deep learning applications demonstrated the clear potential of neural feature extraction, they highlighted specific data-handling challenges inherent to continuous passive monitoring. The windowing methodology utilized to expand training sample sizes relied on slicing continuous transits into short, overlapping frames [2]. When subjected to standard random data partitioning, this technique risks introducing temporal autocorrelation across the validation boundary, as identical localized acoustic states appear concurrently in both partitions.

Furthermore, relying on automated Automatic Identification System (AIS) logs for ground-truth assignment introduces localized spatial-temporal alignment lags during a vessel’s Closest Point of Approach (CPA), generating label noise at transit boundaries [2]. This dissertation directly builds upon the insights of the DST framework. Rather than expanding network complexity, this project shifts the focus toward establishing a highly

interpretable classical machine learning pipeline. It implements strict spatial-temporal validation protocols designed to isolate invariant target mechanics from localized ambient fluctuations.

2.3 Feature Space Engineering and Alternative Matrix Representations

An alternative to deep feature extraction involves mapping raw acoustic pressure fields into specialized mathematical manifolds that optimize traditional supervised classifiers. Research leveraging passive sonar arrays demonstrates that transforming raw data into complex-valued representations such as the phone-space cross-spectral density matrix (PCSDM) or the mode-space cross-spectral density matrix (mCSDM) can isolate unique acoustic indicators [3]. Utilizing the absolute values of these matrices as input vectors enables classical maximum-margin estimators and tree-based ensembles to achieve classification accuracies exceeding 95% when distinguishing surface vessels from underwater targets [3].

This methodology underscores the utility of targeted feature engineering in passive acoustics. Computing spectral density matrices or structural sub-band energy profiles captures the physical characteristics of acoustic sources, bypassing the computational overhead of highly parameterized physical propagation models [3]. This aligns with de Lautour and Liu’s exploration of alternative time-frequency representations, which noted that transforms with logarithmic frequency axes such as the Constant-Q Transform (CQT) yield superior tonal resolution for rotating machinery noise compared to the linear scaling of the standard Short-Time Fourier Transform (STFT) [2].

However, high reported accuracies ($> 95\%$) often stem from simulated environments parameterised with static sound speed profiles and idealized bottom bathymetry [3]. In active maritime deployments, real-world hydrophone channels suffer from hydrodynamic drag, sensor bowing, and clock drift that degrade precise phase relationships [3]. Consequently, rather than relying on brittle phase-coherence matrices, this project extracts a multi-dimensional, tabulated feature space spanning LOFAR tonals, DEMON modulations, and noise-normalized broadband slope metrics. This ensures the engineered feature matrix remains robust against real-world channel degradation.

2.4 Methodological Rigor and Cross-Validation Structures

The structural configuration of the data preprocessing and cross-validation pipeline dictates a model’s stability when transitioning from static validation frameworks to live operational deployment. Historical investigations utilizing standard sonar reference matrices emphasize the necessity of systematic hyperparameter optimization via grid-search algorithms, identifying the traditional Support Vector Machine (SVM) configured with a Radial Basis Function (RBF) kernel as a highly effective model for binary acoustic separation [3].

However, standard benchmarking workflows in tabular data science often execute data scaling, polynomial feature expansion, and dimensionality transformations globally across the complete data matrix prior to partitioning the validation folds [10]. This improper sequencing induces catastrophic distribution leakage, as information regarding the global mean, variance, and feature boundaries of the testing partitions is inadvertently integrated into the training phase [10]. This architectural risk is severely exacerbated in passive marine surveillance, where brief, independent active sonar returns are replaced by continuous, highly auto-correlated passive time-series streams where adjacent windows are bound by

intense sequential dependencies [10].

When sequence data is subjected to standard randomized sample-level splitting, the independent and identically distributed (i.i.d.) assumptions completely break down, leading to severe target leakage and artificially inflated performance profiles [20]. Modern robust multi-scale training paradigms emphasize that sequential time-series classification under non-stationary ambient noise requires validation architectures that explicitly protect model metrics from localized temporal bleeding and structured label drift [20].

To ensure absolute statistical integrity across the engineering lifecycle of this project, we enforce strict task boundaries and a chronological day-based validation block split. By completely separating the data matrix along calendar days, each 24-hour block is treated as an entirely independent environmental domain. Furthermore, all feature pruning, scaling, and imputation parameters are computed exclusively within the training folds. This strict structure protects the pipeline from distribution leakage, ensuring the resulting performance profiles reflect genuine cross-environmental generalization.

Chapter 3

Data Description and Exploration

This chapter provides a detailed explanation and exploratory analysis of the underwater acoustic dataset, outlining the methodologies employed to transition from continuous, unstructured waveforms to a structured, tabulated format suitable for supervised machine learning. The raw acoustic data utilized across this investigation is derived directly from the continuous acoustic recordings captured in previous works [2]. To convert these continuous multi-node audio streams into discrete, high-dimensional observation vectors, the data was processed and tabulated using the specialized automated pipeline developed by Defence Science and Technology (DST) and engineered by Sohan Ghodla. The following sections detail the structural parameters of the survey framework, the mathematical mechanics of the feature extraction domains, and the physical constraints governing the resulting feature matrix.

3.1 The Tamaki Strait Survey Framework

The primary data source for this project is an unclassified underwater noise survey dataset collected from the Rangitoto Channel in Auckland’s Hauraki Gulf, provided by Defence Science and Technology (DST). This baseline monitoring framework leverages the raw physical acoustic recordings captured during the historical deployment operations documented in previous works [2]. To construct the monitoring array, three discrete Soundtrap 300HF underwater sound recorders were bottom-mounted at critical maritime junctions to monitor the radiated noise profiles of passing surface vessels between 20 February and 4 March 2020 [2]. The spatial network arrangement and hardware context are captured in Figure 3.1:

The raw acoustic streams across all three deployment sites were sampled uniformly at a rate of 16 kHz. The transition from these un-tabulated wave arrays into the multi-dimensional, tabulated predictor space utilized by the downstream machine learning models was executed entirely by the automated DST processing pipeline created by Sohan Ghodla. Before evaluating specific feature blocks generated by this pipeline, four fundamental acoustic processing concepts must be defined:

- **Nyquist Frequency:** The Nyquist frequency represents the highest frequency that can be accurately resolved in a digitally sampled signal without aliasing distortion, defined mathematically as half of the total sampling rate. Because the hydrophone recorders sample data at 16 kHz, the resulting Nyquist limit is exactly 8 kHz. This range is sufficient for capturing the essential acoustic emissions associated with ship-radiated noise, such as low-to-mid frequency engine rumble and generic broadband hull noise.



Figure 3.1: Rough Locations of hydrophones within the littoral channel during the surveillance logging campaign.

- **Low-Frequency Analysis and Recording (LOFAR):** LOFAR is a spectral analysis method used to isolate stable, narrowband frequencies, commonly referred to as discrete tonals. These tonals originate from internal rotating machinery, such as propulsion engines, auxiliary generators, and pumps. They manifest as distinct peaks in the power spectrum and can remain detectable over long propagation distances.
- **Detection of Envelope Modulation on Noise (DEMON):** DEMON processing extracts low-frequency modulation periodicities that are embedded within high-frequency broadband noise fields. By tracking the amplitude envelope of a bandpassed acoustic signal, this technique helps uncover cyclic machinery rates, such as shaft rotational speeds or structural acoustic rhythms.
- **Mel-Frequency Cepstral Coefficients (MFCCs):** MFCCs compress the acoustic spectrum by warping frequency distributions onto the logarithmic Mel scale, projecting the resulting energy profile onto a compact set of orthogonal cepstral vectors. This provides a generalised texture descriptor that captures underlying spectral resonance trends.

To transition from raw multi-node waveform arrays to a structured layout suitable for classical machine learning, the synchronised audio fields were processed into a consolidated, tabulated feature set. This processed space maps structural acoustic characteristics, focusing primarily on generic broadband noise measurements and discrete narrow-band tonals, which are the main physical properties leveraged by the downstream machine learning models.

3.2 Target Labelling and Feature Extraction Architecture

Establishing an accurate ground truth remains a primary bottleneck in underwater acoustic target recognition, necessitating the fusion of historical Automatic Identification System

(AIS) spatial telemetry with raw, unclassified acoustic recordings. While AIS provides high-fidelity, geolocated spatial tracks for commercial, industrial, and military vessels, relying exclusively on radio transponder telemetry introduces severe selection biases. Smaller recreational watercraft, non-cooperative vessels, and actors engaged in illicit, unregulated, or unreported (IUU) fishing activities are frequently not mandated to transmit AIS data, or they may intentionally disable, spoof, or manipulate their transponders [8].

To mitigate these telemetry blind spots and ensure a robust validation environment, the data processing pipeline implements a hybrid spatial-acoustic rule-based labelling architecture. This framework cross-references a binary spatial tracking mask (M_{AIS}) against two independent acoustic verification metrics: narrow-band acoustic tonal peak counts (N_{peaks}) and a normalized broadband proxy Signal-to-Noise Ratio (A_{norm}). By checking telemetry data against real-world physical acoustic emissions, the system can reliably isolate true target encounters from complex background noise variations, while simultaneously logging dark targets that lack matching transponder records.

3.2.1 Automated Labelling Heuristics and Sample Weighting Profiles

The dataset labelling strategy integrates these tracking and signal components to classify individual 60-second acoustic chunks, dynamically generating binary labels alongside continuous confidence weights $w \in (0, 1]$. Because tracking telemetry and raw acoustics suffer from localized spatial-temporal alignment lags during a vessel’s Closest Point of Approach (CPA), treating all derived targets as globally deterministic indicators would introduce significant label noise into the training matrix. Instead, target labels and continuous sample weights are updated using a combined spatial-acoustic activity score, balancing historical tracking records with observable physical signal attributes.

The filtering, telemetry alignment, and proxy generation phases are executed across two distinct processing stages:

1. **AIS Spatial Filtering and State Disambiguation:** For each acoustic chunk midpoint (t_{mid}), historical AIS tracking paths are queried within a tight temporal window of ± 50 seconds. This restrictive windowing allows the pipeline to compute instantaneous ping intervals, filtering out stationary, moored, or anchored vessels that do not contribute active machinery or propulsion signatures to the channel. Valid dynamic tracks within a bounding box centered on the active hydrophone station’s geographic coordinates are interpolated onto a fine 2-second grid. This interpolation resolves temporal latency discrepancies, allowing the system to compute the exact range to the target at its closest point of approach ($R_{\text{max}} \leq 4.0$ km). True over-ground mean speed (v_{mean}) and aggregate spatial displacement (d_{disp}) are calculated using generalized Haversine formulations to confirm target motion. If a dynamic vessel track falls within the R_{max} perimeter, the spatial mask is set to $M_{\text{AIS}} = 1$; otherwise, it is designated as $M_{\text{AIS}} = 0$.
2. **Orthogonal Acoustic Proxy Generation:** To capture anthropogenic acoustic energy independent of cooperative radio transmissions, a proxy metric is extracted directly from the chunk’s Low-Frequency Detection and Ranging (LOFAR) Power Spectral Density (PSD). The acoustic proxy signal-to-noise ratio (P_{SNR}) is evaluated over a targeted sub-band spanning 25 Hz to 1000 Hz, which encompasses the primary low-frequency machinery line spectra and broadband propeller cavitation signatures of surface vessels. To protect the proxy from transient ambient inflation, the active chunk’s spectral energy is measured against a historical noise floor derived from the 15th percentile of a rolling 10-minute spectral buffer ($\mathcal{B}$). This background

normalization filters out brief biological clicks and hydrodynamic transients. The resulting proxy value is scaled to a normalized continuous metric $A_{\text{norm}} \in [0, 1]$ and validated against an operational decision threshold (τ_{proxy}).

Following the completion of this hybrid verification loop, the processing pipeline maps the acoustic workspace across seven distinct processing domains. This maps a multi-dimensional acoustic fingerprint designed to isolate invariant structural and mechanical characteristics. While the underlying mathematical equations, windowing transforms, and frame-level formalisms for these extraction domains are detailed in Appendix A, their physical and operational roles within the classifier framework are structured as follows:

- **LOFAR Features (Tonal & Narrowband Signatures):** This domain isolates stable, continuous narrow-band frequencies originating from internal rotating machinery components, such as propulsion engines, auxiliary generators, and pumps. By transforming the continuous waveform into the spectral domain using a median-averaged Welch estimator, the pipeline tracks stable line spectra that remain detectable over long propagation distances. It calculates total narrow-band signal strength, identifies the dominant harmonic configurations, and measures peak configurations to estimate fundamental engine firing rates (f_0), providing a clear descriptor of mechanical classification attributes.
- **DEMON Features (Modulation Signatures):** This processing block isolates low-frequency modulation periodicities that are embedded within high-frequency broadband noise carriers. Driven primarily by propeller cavitation dynamics and blade-rate rotations, DEMON processing isolates the high-frequency carrier band and tracks its amplitude envelope. This temporal envelope is analyzed in the modulation frequency space to uncover cyclic machinery rhythms, shaft rotational speeds, and blade-count periodicities, capturing physical signatures that are highly resilient to shallow-water channel degradation.
- **Broadband Features (Absolute Energy Metrics):** This block quantifies the un-whitened, absolute acoustic energy across the entire sensor bandwidth, segmenting the power distribution into four distinct sub-bands (Flow, Low, Mid, and High frequencies). By measuring raw energy variations, these features capture overall spectral tilt indicators and log-frequency energy decay slopes, serving as a direct operational proxy for target proximity, speed over ground, and physical vessel displacement.
- **Noise-Normalized Broadband Features (SNR Shape Metrics):** To protect downstream estimators from volatile environmental shifts caused by wind-driven surface waves, rain, and changing diurnal soundscapes, this domain isolates the structural shape of the whitened spectrum. By dividing the active chunk’s power distribution by the 10-minute historical rolling noise floor, the feature space eliminates localized channel bias. This allows the models to evaluate relative sub-band SNR profiles and whitened slopes based entirely on true target anomalies.
- **Spectral Metrics (Time-Frame Resolution Descriptors):** This domain tracks the structural and statistical stability of individual spectral frames across time, calculating Wiener flatness indicators and normalized Shannon spectral entropy. By evaluating frame-by-frame variability, these metrics distinguish steady, low-entropy continuous anthropogenic emissions from erratic, high-entropy transient environmental events, such as localized snapping shrimp bursts or wave impacts.

- **Mel-Frequency Cepstral Coefficients (MFCCs):** Spanning from `mfcc_1_mean` to `mfcc_13_std`, this block compresses the acoustic spectrum by warping frequency bins onto the logarithmic Mel scale, projecting the energy distribution onto orthogonal cepstral vectors. This provides the machine learning models with an agnostic, low-dimensional textural descriptor that captures latent hull resonances and machinery acoustic profiles.
- **Temporal Features (Waveform Dynamics):** Extracted directly from the raw, untransformed time-series amplitude waveform, this processing block tracks immediate signal dynamics. By logging temporal Zero-Crossing Rates (ZCR), moving Root-Mean-Square (RMS) power fluctuations, global waveform crest factors, and temporal modulation depths, these features preserve immediate signal characteristics and transient impulses that are typically lost during spectral frequency-domain transformation.

3.2.2 Data Leakage and Circular Logic Risks

A methodological constraint within this framework involves the structural dependency between the automated labelling heuristics and the acoustic feature space. Because the binary target labels are partially generated via rule-based heuristics that utilise acoustic criteria specifically tonal peak counts and broadband proxy signal-to-noise ratios there is a systemic risk of data leakage.

If a classification model is trained on acoustic features that directly correlate with the thresholds used in the labelling rules, the algorithm risks learning the mathematical boundaries of the labelling heuristics rather than the true underlying physical signatures of the vessels. This potential circular logic constraint means the model could end up reversing the rule-based engineering logic instead of identifying invariant acoustic mechanics. To ensure generalisation and protect model integrity, features that directly inform the target assignment rules must be identified and permanently removed prior to the training phase.

3.3 Exploratory Data Analysis

3.3.1 Dataset Metrics and Structural Overview

To establish a benchmark for classical machine learning evaluation, the compiled dataset was structured into a consolidated matrix comprising 114,958 individual observations (rows) across 177 continuous and categorical variables. The survey captured continuous acoustic fields across multiple parallel monitoring stations.

The total cumulative duration of all processed audio chunks across the three hydrophone deployments equals 6,897,480.0 seconds, which scales to approximately 1,915.97 hours or 79.83 sensor-days. This cumulative audio duration exceeds the absolute wall-clock duration because data streams were captured concurrently across the spatial node topology. The overall wall-clock elapsed time span for the survey covers 1,240,023.0 seconds, representing approximately 344.45 hours or 14.35 calendar days.

The signal processing pipeline limits the frequency bands based on the physical limits of the sensors and the acoustic properties of the targets. The structural frequency boundaries implemented across the primary extraction domains are defined as follows:

- **LOFAR Bandwidth Coverage:** Evaluated continuously from 15.0 Hz to 2,000.0 Hz to capture low-frequency machinery tonals and structural propulsion components.

- **DEMON Demodulation Carrier Band:** Set from 1,000.0 Hz to 7,500.0 Hz to isolate high-frequency acoustic modulations.
- **DEMON Modulation Frequency Analysis Space:** Bound from 1.0 Hz to 70.0 Hz to resolve cyclic machinery patterns and shaft rotational dynamics.

A summary of these structural dataset parameters is presented in Table 3.1.

Table 3.1: Foundational Acoustic Survey Dataset Metrics

Dataset Parameter	Value
Total Observations (Chunks)	114,958
Total Extracted Features	177
Total Cumulative Audio Duration	6,897,480.0 s (1,915.97 h)
Overall Wall-Clock Time Span	1,240,023.0 s (14.35 days)
LOFAR Processing Bandwidth	15.0 Hz – 2,000.0 Hz
DEMON Demodulation Carrier Band	1,000.0 Hz – 7,500.0 Hz
DEMON Modulation Analysis Range	1.0 Hz – 70.0 Hz

3.3.2 Class Imbalance and Sub-Heuristic Distribution Analysis

At the macro level, the dataset exhibits a balanced distribution between the binary classes. Class 0, representing pure ambient background noise, contains 58,353 samples, which accounts for 50.76% of the dataset. Class 1, representing active vessel encounters, contains 56,605 samples, accounting for 49.24% of the total data. The resulting background-to-vessel imbalance ratio is 1.03:1. This macro-level balance ensures that global accuracy metrics remain stable indicators of model performance without requiring synthetic oversampling techniques.

However, an evaluation of the underlying target heuristic categories reveals a more complex distribution of acoustic and spatial states. While the binary target vector marks 56,605 observations as positive vessel encounters, these entries are divided among several distinct physical conditions.

The baseline ambient background category contains 44,492 samples (38.70%), representing clear intervals without acoustic or spatial target activity. Genuinely verified vessel encounters with standard acoustic alignments account for 29,462 samples (25.63%). Faint acoustic profiles that display observable narrow-band tonals without matching AIS confirmation comprise 15,889 entries (13.82%).

An operational challenge is presented by the 13,861 samples (12.06%) where an AIS track is spatially validated within the 4.0 km radius, yet the vessel remains acoustically silent. Conversely, uncooperative or untracked dark vessels which generate strong acoustic profiles but lack any matching AIS telemetry account for 8,565 samples (7.45%).

The remaining segments consist of 2,039 samples (1.77%) of confirmed vessel positions that display no discrete narrow-band tonals, and 650 samples (0.57%) where acoustic energy thresholds are triggered in the absence of any tracking data. This detailed sub-heuristic breakdown is structured in Table 3.2.

3.3.3 Temporal Auto-Correlation and Statistical Dependency

Because the hydrophone stations continuously record stationary and passing sources, the extracted feature series displays a high degree of temporal auto-correlation. A vessel transit across a sensor’s detection radius is a continuous physical event that spans multiple

Table 3.2: Distribution of Target Labelling Heuristic Sub-Reasons

Heuristic Label Category	Sample Count	Percentage (%)
Ambient Background Noise	44,492	38.70
Standard Verified Vessel Encounter	29,462	25.63
Faint Acoustic Tonals Only	15,889	13.82
AIS Present (Acoustically Silent Vessel)	13,861	12.06
Untracked Target (Dark Vessel Signature)	8,565	7.45
Confirmed Vessel Position (No Machinery Tonals)	2,039	1.77
Acoustic Threshold Trigger (No Transponder Data)	650	0.57

consecutive time windows. Consequently, the acoustic properties of a given chunk X_t are highly dependent on the preceding states $X_{t-1}, X_{t-2}, \dots, X_{t-k}$.

This temporal auto-correlation can be modeled statistically. The conditional probability of a positive vessel label given a positive label in the immediate prior interval approaches a high probability:

$$P(Y_t = 1 \mid Y_{t-1} = 1) \gg P(Y_t = 1) \quad (3.1)$$

If this temporal dependency is ignored during validation design, it introduces a major risk of distribution leakage. Standard random data splitting breaks the sequential order of the observations, placing adjacent windows from the same target passage simultaneously into the training and testing matrices.

The classifier would then train on the localized background noise and specific machinery fluctuations of a transit, artificially inflating its validation scores. When deployed against an independent deployment day or a separate geographic channel, the model’s performance drops because it memorised localized temporal states rather than learning invariant target features. This intense temporal dependency and risk of severe model overfitting mandates the use of blocked structural partitioning frameworks to separate structural features cleanly [10]. This auto-correlation structure justifies the use of a day-separated validation block split, ensuring that evaluation boundaries align with complete, independent diurnal cycles.

3.3.4 Physical Taxonomy of Feature Missingness

An evaluation of the feature matrix reveals structural missingness patterns across both the LOFAR and DEMON processing blocks. This missingness is a direct result of the physical properties of the underwater environment rather than a consequence of hardware telemetry failures, with explicit schemas detailed in Appendix B:

- **Narrowband Tonal Detection Thresholds ($\sim 40.02\%$ Missingness):** Features tracking fundamental narrowband properties, such as the dominant frequency or the first prominent spectral peak, display a baseline missingness of 40.02%. This occurs when ambient noise from tidal motion or biological transients masks weak machinery emissions. If no vessel is operating nearby, the peak-detection algorithms fail to resolve any discrete spectral lines, returning a null value.
- **Harmonic Spacing Constraints ($\sim 48.40\%$ Missingness):** Missingness rises to exactly 48.40% for the mean peak spacing and the coefficient of variation of peak spacing. This step-change is a mathematical property of the feature formulas: calculating a spacing metric requires a minimum of two resolved spectral peaks ($k \geq 2$). If the processing pipeline resolves only a single machinery tonal before

the remainder submerge into the background noise, spacing parameters cannot be calculated and return a null state.

- **Cascading Higher-Order Harmonics (Up to 63.31% Missingness):** The missingness rate increases sequentially when tracking higher-order mechanical harmonics from the third through the sixth spectral peaks. Because spectral peaks are sorted by prominence, higher-order rotational components submerge into the broadband noise floor. This rising null rate reflects the physical reality that few vessel transits maintain six simultaneously observable harmonics in a shallow coastal channel.
- **Modulation Extraction Boundaries ($\sim 93.26\%$ Missingness):** Features tracking frequency-domain modulation metrics, such as the dominant modulation frequency or the estimated fundamental machinery rate, display an extreme missingness rate of 93.26%. This pattern is governed by target velocity limits. If a vessel travels at low speeds, or if its propulsion systems are optimized to limit vibration, no clear amplitude modulation is generated. The resulting modulation spectrum remains flat, preventing the extraction algorithms from resolving cyclic shaft or blade rates.

Chapter 4

Methodology

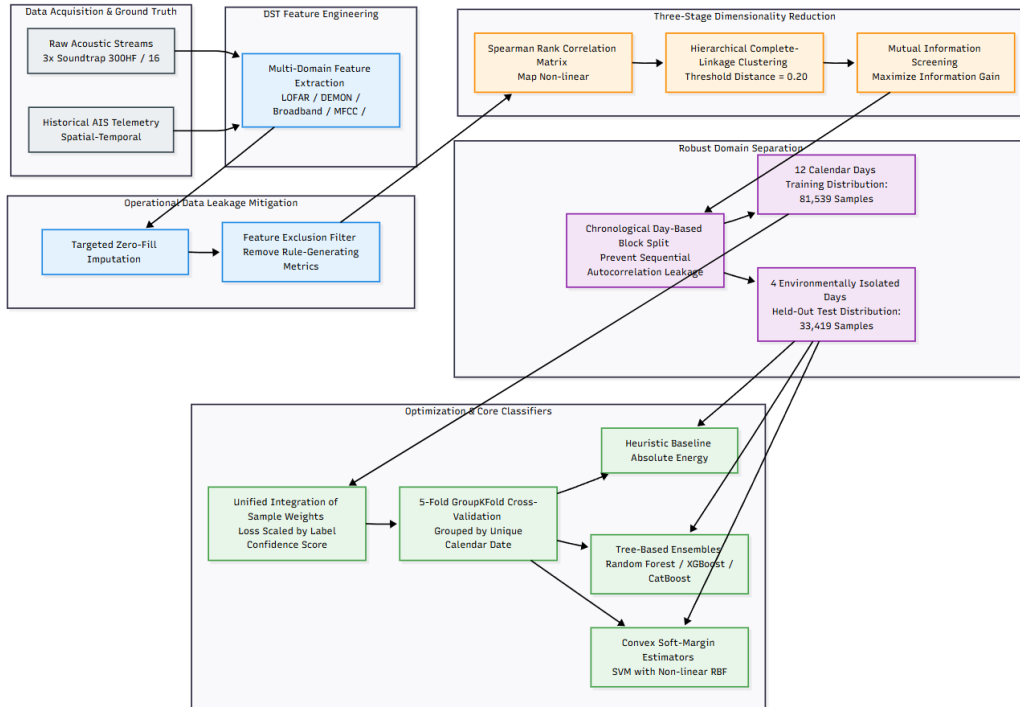


Figure 4.1: Schematic representation of the data ingestion, targeted feature reduction, chronological block cross-validation framework, and confidence-weighted classifier optimization loop used for baseline uncooperative maritime surveillance.

This chapter defines the machine learning frameworks, data preprocessing pipelines, and validation strategies applied to resolve the binary acoustic detection problem, with the high-level functional sequencing illustrated in Figure 4.1. The processing sequence is structured to maximize model generalisation across independent environmental domains while protecting the development lifecycle from mathematical and structural data leakage.

4.1 Mitigation of Operational Data Leakage

4.1.1 Targeted Imputation Strategy

Standard statistical imputation techniques, such as global mean, median, or K-Nearest Neighbors substitution, insert severe structural flaws when applied to acoustic features governed by rigid physical boundaries. Imputing the average value of a machinery tonal frequency into empty observations would falsely simulate the presence of active engines and rotating propeller shafts during intervals of pure ambient noise. This baseline distortion shifts the model decision hyperplanes away from true physical criteria.

Because missingness within the Low-Frequency Analysis and Recording (LOFAR) and Detection of Envelope Modulation on Noise (DEMON) feature blocks represents the physical absence of specific acoustic energy components, a deterministic zero-fill imputation strategy is implemented. Let $\mathbf{X} \in \mathbb{R}^{N \times M}$ represent the un-imputed feature matrix, where $x_{i,j}$ denotes the scalar value of the i -th sample for the j -th acoustic feature. The imputation function is defined as:

$$x_{i,j} = \begin{cases} x_{i,j} & \text{if } x_{i,j} \neq \text{NaN} \\ 0 & \text{if } x_{i,j} = \text{NaN} \end{cases} \quad \forall i \in \{1, \dots, N\}, j \in \{1, \dots, M\} \quad (4.1)$$

This systematic missingness pattern is explicitly classified as Missing Not At Random (MNAR), given that the probability of a feature being logged as null is directly tied to the unobserved value itself. In acoustic automated surveillance, missing fields occur precisely because the signal-to-noise ratio of a specific tonal peak or amplitude envelope drops below the extraction algorithm’s tracking threshold, or because the physical source is entirely absent. Mapping these systematic missing states to a deterministic 0.0 aligns cleanly with acoustic theory; it treats an unresolved peak prominence or shaft modulation parameter as an objective lack of narrow-band or cyclic mechanical energy at that coordinate, preventing mean-driven spectral inflation.

Within the underwater acoustic target recognition domain, this approach addresses a distinct architectural trade-off. Traditionally, standard practice has circumvented tabular data missingness altogether by processing continuous, un-tabulated time-frequency matrix representations such as raw spectrogram images directly through deep neural networks. However, when designing highly interpretable baseline systems that rely on explicit, engineered feature dictionaries derived from automated peak-picking routines, zero-filling serves as a standard, physically intuitive protocol. While modern gradient-boosted ensembles can process raw missing indicators natively, standardizing the null coordinates to zero ensures absolute mathematical compatibility across both non-parametric tree structures and convex soft-margin estimators, allowing the classifiers to isolate structural domain boundaries without mapping artifacts. This approach is highly critical when evaluating data from decentralized multi-node littoral architectures, where un-synchronized sensor placements frequently generate sparse arrays and unresolved channel coordinates due to local fading or localized shadow zones [15]. By mapping the absolute absence of a signal to a distinct, localized numerical state (0.0), the pipeline preserves the underlying physical boundaries of the domain. This structural formatting allows tree-based ensembles and maximum-margin classifiers to partition missing spectral fields cleanly from active target responses across all nodes in the network.

4.1.2 Filtering of Rule-Generating Features

To ensure the statistical integrity of the evaluation metrics, features that directly inform the automated target generation scripts must be isolated and permanently removed. The parameters tracking total tonal peak counts and broadband signal-to-noise ratios, along with their derivative components (e.g., `snr_broadband_db`), were directly utilized by rule-based algorithms to construct the ground-truth target vector.

Retaining these variables in the training matrix would allow the machine learning models to solve the inverse mapping of the labelling heuristics rather than learning the independent physics of ship-radiated noise. This circular logic constraint would produce artificially perfect training performance metrics that fail completely when exposed to un-labelled field data. An automated string-matching routine identified and excluded these metrics, reducing the input matrix to a cleansed set of features that represent only the un-leaked physical characteristics of the target encounters.

4.1.3 Chronological Day-Based Validation Block Split

Acoustic waveforms collected via continuous passive monitoring exhibit intense temporal dependencies. Consecutive observation windows share nearly identical physical waveguide attributes, localized noise states, and target propulsion dynamics. Applying standard randomized validation partitioning or randomized K -fold cross-validation at the sample level would distribute adjacent windows from the same continuous vessel transit simultaneously into both the training and testing matrices. This spatial-temporal data leakage allows the model to memorize the localized ambient background states of specific encounters, artificially masking its vulnerability to cross-domain generalization collapse [1, 10].

To mitigate this operational bias, a chronological day-based block split strategy is deployed. The data matrix is explicitly partitioned along calendar days, treating each discrete 24-hour block as an independent environmental domain. This structure forces a true cross-environmental stress test, breaking the localized sequential bleeding of temporal noise floors [10]. Rather than deploying a randomized split on the days, an operational held-out testing distribution is explicitly defined. Four discrete days are selected as the permanent validation baseline: February 20, February 24, February 28, and March 3, 2020.

The selection of this specific testing subset is fundamentally anchored by the fact that these four discrete calendar days were subjected to rigorous manual auditing, annotation, and verification by DST researchers. While the broader training matrix relies on automated, rule-based algorithmic heuristics that inherit localized telemetry and proxy alignment noise, these specific validation blocks were thoroughly cross-examined and reviewed to isolate true vessel profiles from ambient anomalies. Consequently, barring human error or undetected hardware telemetry dropouts, these days establish the most definitive and high-fidelity ground-truth baseline available to validate model performance. Structurally, their selection is further justified by distinct environmental and maritime traffic conditions captured across the survey timeline:

- **February 20 and February 28, 2020:** These dates capture high-density commercial shipping traffic during weekday operational peaks. They provide complex multi-source acoustic profiles and clear machinery line spectra within the channel.
- **February 24 and March 3, 2020:** These dates capture shifting environmental boundaries. They are characterized by localized wind-driven surface waves and tidal flow variations that elevate the continuous low-frequency noise floor, balanced against low recreational vessel traffic windows.

This structured distribution forces the validation partition to serve as a comprehensive cross-environmental test bed. The remaining twelve calendar days comprise the training partition distribution, yielding 81,539 training observations and 33,419 domain-isolated testing observations.

4.2 Feature Pruning & Correlation Analysis

Following the exclusion of features tied to target creation, the raw input matrix contains 148 columns. Due to parallel extraction processes across the spectral processing blocks, this feature space displays high multicollinearity, which destabilizes maximum-margin estimators and increases tree-split variance. To reduce dimensionality while preserving physical interpretability, a clustering-based feature selection routine is executed via a three-stage mathematical filter.

First, to capture non-linear but monotonic relationships that standard linear metrics miss, the pipeline calculates the pairwise Spearman rank correlation matrix $\mathbf{R}_\rho$ for all features on the training set. The Spearman coefficient ρ_{ij} is computed as:

$$\rho_{ij} = 1 - \frac{6 \sum_{k=1}^n d_k^2}{n(n^2 - 1)} \quad (4.2)$$

where d_k is the difference between the ranks of the corresponding variables for sample k , and n is the total number of samples within the training partition.

Second, the correlation matrix is transformed into a non-linear cophenetic distance matrix $\mathbf{D}$, where elements are defined as $d_{ij} = 1 - |\rho_{ij}|$. Agglomerative hierarchical clustering is applied to this distance space using complete-linkage criteria, bounding the cluster boundaries with a distance threshold of $\tau = 0.20$. This threshold implies that any features grouped within a unified hierarchical cluster share a minimum absolute rank correlation of $|\rho_{ij}| \geq 0.80$.

Third, to resolve redundancy within each discovered cluster without losing predictive power, the pipeline screens features using an information-theoretic criterion. For each independent feature X_j within a cluster, the Mutual Information (MI) score relative to the discrete target variable Y is calculated:

$$I(X_j; Y) = \sum_{x \in \mathcal{X}_j} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (4.3)$$

where $p(x, y)$ is the joint probability distribution function of X_j and Y , and $p(x)$ and $p(y)$ are the respective marginal probability distribution functions. The feature that maximizes $I(X_j; Y)$ is selected as the optimal representative for that cluster, while the remaining collinear parameters are permanently pruned. This strategy explicitly minimizes target variable redundancy while maximizing individual feature dependency, mapping directly to the optimization priorities of the Max-Relevance Min-Redundancy (mRMR) structural selection framework [7].

This routine compresses the feature space from 148 raw metrics down to 76 optimized acoustic predictors. While the Spearman clustering pipeline removes 64 redundant variables, a dimensionality of 76 features remains high for classical architectures deployed in real-time embedded environments. Advanced refinement techniques such as permutation feature importance metrics or greedy forward beam selection are critical areas for future work to further streamline the operational feature space.

4.3 Model Selection & Core Architectures

To evaluate the trade-offs between model complexity, computational efficiency, and predictive robustness within a shallow-water waveguide, five diverse classifiers were selected for baseline benchmarking.

4.3.1 Decision Trees

The Decision Tree serves as the baseline, non-parametric structural block. For this binary classification problem, it recursively partitions the acoustic feature space by maximizing the reduction of Shannon entropy. The entropy of a given node split T is defined as:

$$H(T) = - \sum_{i=1}^C p_i \log_2 p_i \quad (4.4)$$

where $C = 2$ represents the binary classes (noise versus vessel), and p_i is the probability of an observation belonging to class i within that node. This formula measures information gain via Shannon entropy, which quantifies the reduction in spectral uncertainty, distinguishing it from the Gini impurity index that minimizes misclassification probability.

Project-specific reasoning justifies this model as a baseline because it provides clear, interpretable hierarchical thresholds for distinguishing loud vessel transits from quiet ambient periods. Regarding model vulnerabilities, because the feature extraction windows are 60 seconds wide, transient acoustic pops (such as isolated biological clicks lasting fractions of a second) do not cause localized tree overfitting. Instead, persistent snapping shrimp activity elevates [2] the continuous broadband noise floor across the entire processing window. This continuous elevation shifts the distribution of energy-based splits, leading to structural instability in standalone shallow trees.

4.3.2 Random Forests

The Random Forest addresses individual tree instability through bootstrap aggregating (bagging), constructing an ensemble of B independent decision trees, denoted as $f_b(x)$. The final binary classification decision is determined via a majority vote of the ensemble:

$$\hat{f}_{\text{RF}}^B(x) = \arg \max_{c \in \{0,1\}} \sum_{b=1}^B \mathbb{I}(f_b(x) = c) \quad (4.5)$$

where $f_b(x)$ is the class prediction of the b -th individual tree, $\mathbb{I}(\cdot)$ is the indicator function that outputs 1 if the interior conditional statement is true and 0 otherwise, c is the binary class variable, and the $\arg \max$ operator selects the class index that maximizes the aggregated sum.

This ensemble model fits the acoustic dataset because random subspace sampling evaluates only a small subset of $m \approx \sqrt{p}$ features at each individual node split. This mechanism decouples collinear features locally, preventing dominant variables (such as a specific broadband energy band) from masking more subtle, highly discriminative indicators like narrowband spectral slope parameters during tree construction. However, at the final ensemble layer, the predictions of these independent trees are combined via majority voting. This aggregation process effectively reduces structural prediction variance across changing ambient conditions, but it averages the outputs rather than fundamentally altering or removing the underlying multicollinearity present in the cumulative joint feature matrix.

4.3.3 Gradient Boosting (XGBoost and CatBoost)

Gradient Boosting frameworks build ensembles sequentially via an additive training process. The initial base learner is fit directly to the raw target data, while each subsequent tree $f_t(x)$ is trained to fit the residual errors or negative gradients generated by the prior ensemble’s joint predictions.

To protect the model from non-stationary noise variations, the optimization routine minimizes a regularised training objective function $\mathcal{L}^{(t)}$ at step t , which is distinct from the per-sample loss function ℓ :

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \ell \left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i) \right) + \Omega(f_t) \quad (4.6)$$

where ℓ is the differentiable per-sample loss function measuring the error between the true binary label $y_i \in \{0, 1\}$ and the model prediction $\hat{y}_i$. The term $\Omega(f_t)$ represents the structural regularization function that penalizes model complexity:

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 + \alpha \sum_{j=1}^T |w_j| \quad (4.7)$$

where T is the total number of terminal leaves, w_j is the real-valued score assigned to leaf j , λ represents the L_2 regularization penalty hyperparameter, and α represents the L_1 regularization penalty hyperparameter. This regularised framework balances structural bias and variance when evaluating complex, time-varying acoustic spectrum descriptors.

While standard implementations like XGBoost optimize structural splits effectively, they suffer from a theoretical limitation when deployed against continuous passive monitoring series. Because adjacent 60-second feature windows share highly correlated background noise states and propagation characteristics, standard boosting frameworks introduce target leakage during tree construction. This leakage triggers an operational phenomenon known as prediction shift: the distribution of calculated gradients on the training data shifts systematically relative to the true distribution encountered on independent validation domains. Mathematically, the gradient computed for an individual sample x_i at step t utilizes a model $\hat{y}^{(t-1)}$ that was trained using that same sample’s target label y_i . This conditional dependence breaks the assumption of independent and identically distributed (i.e.d.) observations, causing the model to overfit to localized temporal sequences instead of generalizing across independent days.

CatBoost structurally counteracts this prediction shift by implementing an Ordered Boosting paradigm. Let $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ represent a random permutation of the training dataset index space. To compute the un-biased residual gradient for a specific observation x_i , CatBoost maintains an independent ensemble model M_i that is trained exclusively on a subset of samples that precede x_i within the permutation sequence, satisfying $\sigma(j) < \sigma(i)$. The gradient evaluation for x_i is defined as:

$$g_t(x_i, y_i) = \left. \frac{\partial \ell(y_i, s)}{\partial s} \right|_{s=M_i(x_i)} \quad (4.8)$$

By decoupling the sample evaluation from the model’s structural update history, the sequential feedback loop of adjacent auto-correlated timeframes is broken.

Furthermore, CatBoost alters the underlying tree geometry by utilizing symmetric or oblivious trees, where identical splitting criteria (same feature and same threshold) are applied uniformly across all nodes at a given depth level. The index of a terminal leaf can

be evaluated via a sequence of binary operations, mapping directly to bitwise execution commands:

$$\text{Leaf Index} = \sum_{d=0}^{D-1} \mathbb{I}(x_{j_d} > \tau_d) \cdot 2^d \quad (4.9)$$

where D is the tree depth, j_d is the feature index evaluated at depth level d , and τ_d is the scalar split threshold. Oblivious structures act as a powerful regularizer; because the splitting decisions are balanced and globally constrained, the model avoids isolating high-variance, overfit pathways driven by short-term biological anomalies like snapping shrimp bursts. When continuous features are quantized into discrete levels or target statistics are calculated, this structural symmetry smooths out sudden covariate shifts caused by changing diurnal biological soundscapes, allowing CatBoost to achieve high cross-environmental stability on day-separated validation blocks.

4.3.4 Support Vector Machines

The Support Vector Machine (SVM) offers a convex alternative to tree-based architectures. Mapping data into an infinite-dimensional Hilbert space is entirely kernel-dependent, achieved here by implementing a non-linear Radial Basis Function (RBF) kernel, rather than being an inherent default property of standard linear SVMs. The RBF kernel function is expressed as:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (4.10)$$

where $\gamma > 0$ is the hyperparameter controlling the radius of influence of individual support vectors.

The algorithm establishes a soft-margin hyperplane that balances geometric separation against controlled misclassifications under low signal-to-noise ratio (SNR) conditions. The optimization objective is formulated as:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (4.11)$$

where w is the weight vector orthogonal to the separating hyperplane, b is the bias scalar shift, ξ_i is the non-negative slack variable measuring the margin violation distance for observation i , and $C > 0$ is the regularization hyperparameter that scales the penalty incurred by margin violations. The parameter $y_i \in \{-1, 1\}$ represents the adjusted binary target class, and $\Phi(x_i)$ is the non-linear mapping function defined implicitly by the RBF kernel. This soft-margin framework allows the model to handle overlapping spectral feature boundaries without destabilizing its maximum-margin boundaries.

4.4 Hyperparameter Tuning Framework

To maximize the cross-environmental generalization capacity of the core architectures, a systematic hyperparameter tuning framework is integrated directly into the training loop. To prevent temporal distribution leakage, a 5-fold `GroupKFold` validation mechanism is enforced across all optimization operations. This protocol guarantees that all training observations belonging to a specific calendar date are bound exclusively to a single cross-validation fold, forcing the models to predict acoustic profiles on completely unobserved days.

The baseline Decision Tree hyperparameter space is evaluated using an exhaustive grid search (`GridSearchCV`) to locate the exact global parameters for simpler rule boundaries.

Conversely, for high-dimensional ensemble spaces like Random Forests and XGBoost, an exhaustive grid search is computationally restrictive. To maintain resource efficiency, the `RandomizedSearchCV` framework is implemented: instead of evaluating every single combination across a rigid grid, the algorithm samples the parameter space efficiently by randomly drawing a pre-specified number of parameter configurations from defined probability distributions. This randomized sampling provides a high probability of identifying an optimal parameter configuration within a fraction of the compute budget.

4.4.1 Unified Integration of Label Confidence Weights

The target labelling heuristics introduce varying degrees of spatial and acoustic uncertainty based on distance boundaries and sensor placement. Automated ground-truth structures built on spatial AIS parameters can encounter dynamic synchronization drift, transceiver latency, and label degradation at vessel boundaries. Rather than treating all derived targets as globally deterministic indicators, the framework maps continuous confidence scores ($w_i \in (0, 1]$) directly into the optimization loss function via instance sample-weight parameter vectors (`sample_weight`), forcing the downstream estimators to map invariant physical mechanics rather than heuristic label boundary noise [9].

In modern machine learning theory, learning from data with label noise requires rigorous regularization to prevent highly parameterized models from memorizing corrupted or misaligned targets near decision boundaries [17]. To handle this type of instance-dependent and asymmetric label noise, recent architectures deploy per-sample Dirichlet weighting vectors derived from label transition matrices, allowing estimators to down-weight uncertain transitions dynamically [16]. By implementing continuous sample weight parameters, the training lifecycle transitions away from empirical risk minimization toward a conditional distributionally robust optimization paradigm [18]. This framework enforces the mathematical separation of true physical anomalies from noisy, heuristic boundaries by structurally bounding the risk associated with ambiguous, borderline target assignments [18].

The sample weight vector modifies the underlying loss formulations across the models as follows:

- **In Tree-Based Boosters and Forests:** The scalar weight w_i directly scales the gradients and Hessians calculated for each observation in XGBoost, and scales the bootstrap selection probabilities and internal information gain metrics during Random Forest node partitioning. Errors on high-confidence target passes trigger large updates, while noisy, ambiguous observations are suppressed.
- **In Support Vector Machines:** The confidence metric scales the soft-margin penalty hyperparameter C individually for each observation, modifying the penalty sum to $\sum C \cdot w_i \xi_i$. Consequently, margin violations on low-confidence segments incur a lower loss penalty, forcing the optimization path to prioritize the geometric boundaries of clear, verified target signals [9].

Chapter 5

Model Development, Evaluation & Error Analysis

This chapter presents the empirical results, evaluation metrics, and error classifications for the five machine learning architectures trained under the day-separated validation protocol. The performance of these models is analyzed relative to traditional methods and framed within operational naval deployment constraints.

5.1 Previous CNN vs. Standalone Machine Learning Frameworks

To establish a rigorous evaluation baseline, the current classical machine learning pipeline is benchmarked against the deep learning methodologies documented in previous agency research. The prior investigation explored a pre-trained Wavegram-Logmel-CNN and a custom-designed CNN to execute binary ship-versus-noise classification using the same underlying Rangitoto Channel acoustic recordings, highlighting critical architectural, methodological, and operational trade-offs.

Architecturally, the frameworks represent opposing paradigms. The deep learning approach converts one-dimensional acoustic waveforms into two-dimensional spectrogram images to leverage computer vision techniques. While highly capable of automated feature extraction, these neural networks remain structurally opaque, computationally intensive, and highly parameterized, complicating real-time deployment on edge-hardware. In contrast, this project maps raw acoustic data across specialized signal-processing manifolds: LOFAR narrowband tonals, DEMON blade-rate modulations, absolute broadband energy profiles, and temporal waveform dynamics. A clustering-based feature selection routine compresses this predictor space to 76 physically interpretable variables, creating a transparent, reproducible pipeline optimized for resource-constrained embedded systems.

Methodologically, the validation structures dictate cross-environmental reliability. The previous deep learning framework expanded sample sizes by slicing continuous transits into short, overlapping frames subjected to standard random data partitioning. This technique introduces intense spatial-temporal data leakage across the validation boundary, as adjacent windows from the same passage appear concurrently in both partitions. The classifier then memorizes localized noise floors and specific machinery fluctuations rather than learning invariant physics, artificially inflating validation scores. To eliminate this bias, the current framework enforces a chronological day-based block split, partitioning data along calendar days to treat each 24-hour block as an independent environmental domain. This forces a

genuine cross-environmental stress test against shifting diurnal biological soundscapes and wind-driven noise floors.

Empirically, the classical framework yields superior stability over neural network structures. The pre-trained Wavegram-Logmel-CNN achieved a Class 1 Precision of 1.00 but suffered from a restrictive Class 1 Recall of 0.263, frequently misclassifying active transits as ambient background. The custom CNN configuration enhanced Class 1 Recall to 0.621 and global accuracy to 92.6%, but triggered localized overfitting that required aggressive regularization. Protected from temporal data leakage, the tuned CatBoost and XGBoost models achieve global validation accuracies of 98.27% and 98.41%, respectively. More importantly, by integrating continuous label confidence weights into the optimization loss function to suppress spatial-temporal alignment noise near a vessel’s closest point of approach, the confidence-weighted XGBoost architecture secures a Class 1 Precision of 0.99 and a Class 0 Recall of 0.99, establishing a mathematically sound, operationally reliable surveillance baseline with minimal computational overhead.

5.2 Comprehensive Model Evaluation & Metrics Pivot

While macro-level metrics indicate a globally balanced dataset across the entire 14-day survey window, localized segments of the time-series present micro-level imbalances during operational deployment. Long periods of ambient noise punctuated by transient vessel transits alter the local classification context. Consequently, raw global classification accuracy is downplayed as a primary indicator of performance. Evaluation instead prioritises Precision, Recall, and F1-scores to track target capture rates and false alarm containment across shifting localized data densities.

Due to the cubic computational scaling ($\mathcal{O}(N^3)$) inherent to soft-margin optimization over large training matrices, the Support Vector Machine (SVM) was trained on a 10% stratified random sample of the core training partition, amounting to 8,154 observations. Upon convergence, support vectors constituted 42% of this downsampled training subset (3,425 individual support vectors), indicating a high density of support boundaries within low signal-to-noise ratio (SNR) bands.

The comprehensive validation metrics across all five optimized models, alongside the previous convolution neural network model, are compiled in Table 5.1.

Table 5.1: Comprehensive Validation Metrics and Target Performance Profiles

Classifier	Validation Framework	Accuracy	Class 0 (Noise)		Class 1 (Vessel)	
			Precision	Recall	Precision	Recall
Previous CNN	Heuristic Energy Threshold	0.926	0.998	0.917	0.62	0.99
Decision Tree	Complete Training Matrix	0.9774	0.97	0.98	0.98	0.97
Random Forest	Complete Training Matrix	0.9616	0.98	0.94	0.95	0.98
XGBoost	Complete Training Matrix	0.9841	0.98	0.99	0.99	0.98
XGBoost (Weighted)	Complete Training Matrix	0.9842	0.98	0.99	0.99	0.98
CatBoost	Complete Training Matrix	0.9827	0.98	0.99	0.99	0.98
SVM (RBF Kernel)	10% Stratified Sample Subset	0.95	0.97	0.94	0.94	0.97

5.3 Operational Naval Trade-off Analysis

The empirical performance across CatBoost and the XGBoost architectures requires an operational assessment framed around naval deployment trade-offs. While the confidence-weighted XGBoost model achieves the highest raw global metric validation profile (0.9842), it shares an identical Class 1 Precision (0.99) and Class 0 Recall (0.99) macro profile with both the unweighted XGBoost and CatBoost models.

In maritime surveillance operations, the tactical costs associated with classification errors are highly asymmetric:

- **Tactical Cost of a False Alarm (False Positive):** Triggers high economic and operational penalties. In resource-constrained naval contexts, dispatching reconnaissance assets such as Royal New Zealand Navy Offshore Patrol Vessels (OPVs) or Royal New Zealand Air Force P-8A Poseidon maritime patrol aircraft to investigate a false target contact consumes fixed fuel allowances, accelerates equipment maintenance cycles, and induces crew fatigue. False alarms degrade command trust in automated early-warning networks. Representative spectral and energy indicators of this misclassification mode are visualized in Figure 5.1.
- **Tactical Cost of a Missed Detection (False Negative):** Introduces security vulnerabilities, allowing uncooperative, untracked, or illicit vessels to transit strategic coastal channels or enter restricted marine zones undetected. This failure mode typically occurs when transient structural clicks or overlapping environmental dynamics submerge low-frequency propulsion features, as illustrated in Figure 5.2.

This critical operational trade-off is heavily influenced by the spatiotemporal mismatch between cooperative tracking frameworks (AIS) and non-cooperative passive sensors during a vessel’s Closest Point of Approach (CPA) [19]. Because tracking logs and physical acoustic returns frequently exhibit temporal latency discrepancies and spatial target alignment conflicts near the sensor perimeter, relying on raw telemetry boundaries introduces localized label ambiguity [19]. Resolving these target conflicts requires information-theoretic fusion rules to balance missing transponder telemetry against real-world physical acoustic emissions [19].

For wide-area surveillance or automated screening where secondary verification layers are available, CatBoost or unweighted XGBoost configurations are suitable due to their high Class 1 Recall (0.98), which minimizes missed contacts. However, for direct operational command decisions within resource-constrained maritime sectors, the confidence-weighted XGBoost framework remains recommended. By scaling the optimization loss functions by target verification scores, it penalizes low-confidence samples and stabilizes prediction boundaries against background fluctuations, securing a Class 1 Precision of 0.99. When this system flags an alert, command elements can deploy response assets with high confidence that a verified physical target is present in the channel.

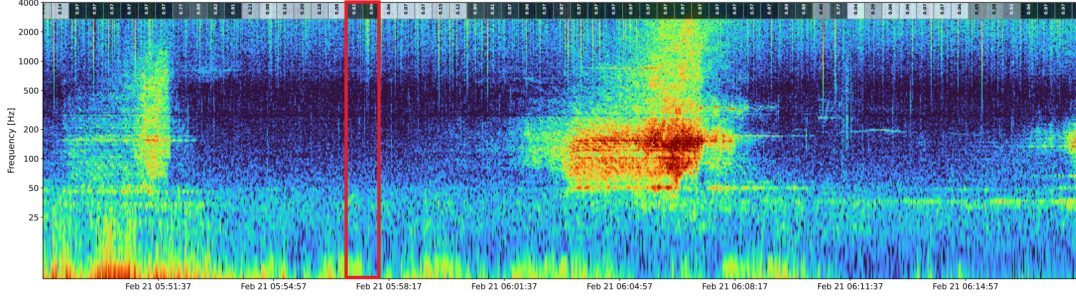


Figure 5.1: False Positive spectrogram. The model detects acoustic energy (ambient noise or non-vessel sound) but misclassifies it as a vessel detection.

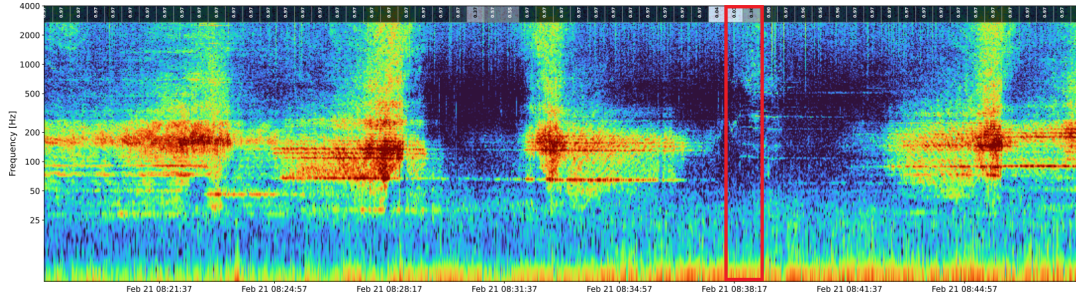


Figure 5.2: False Negative Spectrogram. The model encounters a strong signal indicative of a vessel detection (short burst of patterned energy) but misclassifies it as ambient noise, leading to a missed detection.

5.4 Stratified Confidence-Weighted Error Analysis

To align model evaluation with the confidence-weighted optimization objective used during training, test set performance is stratified across explicit label confidence tiers. Standard unweighted metrics treat all misclassifications uniformly, which introduces evaluation bias

by penalizing models for failing to resolve highly ambiguous, low-confidence samples (e.g., faint machinery acoustic tonals or acoustically silent vessels with spatial AIS-only presence).

By evaluating the loss-aligned weighted error rate across distinct confidence bins, we isolate a model’s true performance on clean, deterministic targets from its capacity to absorb variance along noisy heuristic boundaries. The weighted error rate for each individual confidence bin is calculated mathematically as:

$$\text{Weighted Error Rate} = \frac{\sum_{i=1}^{N_b} (w_i \cdot e_i)}{\sum_{i=1}^{N_b} w_i} \quad (5.1)$$

where N_b represents the total number of observations within the given confidence bin, w_i denotes the continuous sample confidence weight ($w_i \in (0, 1]$), and $e_i \in \{0, 1\}$ is an indicator variable mapping a discrete classification error.

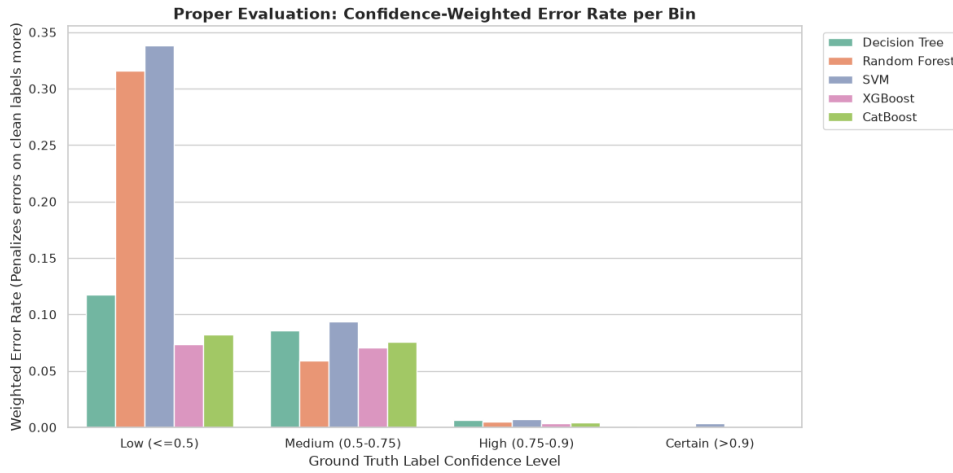


Figure 5.3: Confidence-stratified weighted error rate profiles evaluated across the low, medium, high, and certain ground-truth tier boundaries.

The empirical tracking profiles across the confidence tiers are illustrated in Figure 5.3. An analysis of the resulting distributions indicates that the highest error rates across almost all evaluated models are concentrated within the low-confidence tier (≤ 0.5), followed by a steady, monotonic decrease as label certainty increases. This concentration is a direct reflection of the physical and heuristic parameters governing the low-confidence bin, which encapsulates borderline acoustic returns near the threshold of sensor detection.

Within this low-confidence sector, the Support Vector Machine (SVM) and Random Forest architectures exhibit the highest error rates. The SVM struggles in this region because low-confidence target boundaries introduce significant class overlap in the high-dimensional feature space; this non-separable distribution forces the soft-margin optimizer to accumulate large slack variables (ξ_i) and inflates the density of support vectors near noisy boundaries, degrading its global separating hyperplane. Concurrently, the Random Forest lags here due to its random subspace sampling mechanism: when constructing splits on high-uncertainty observations, individual trees frequently sample weak or unwhitened noise-filled features rather than stable mechanical line spectra, driving up the aggregate ensemble voting variance.

Conversely, within the medium-confidence tier ($0.5 \leq w_i \leq 0.75$), the Random Forest model exhibits a sharp performance recovery, drastically decreasing its weighted error rate to become the top-performing model in this specific sector. This rapid stabilization occurs

because as the underlying target signatures transition from obscure noise boundaries to moderately distinct acoustic encounters, the bootstrap aggregating (bagging) framework of the Random Forest becomes highly protective. By averaging predictions across multiple uncorrelated decision trees trained on distinct data patches, the ensemble effectively dampens localized environmental variance and minor temporal fluctuations, outperforming the sequential gradient-boosting models which can remain overly sensitive to residual alignment noise during early training iterations.

5.5 Non-Parametric Bootstrap Performance and Confidence Interval Analysis

Standard point-estimate evaluations on a static test partition fail to quantify sampling uncertainty or demonstrate whether observed performance variances represent true algorithmic distinctions. To resolve this statistical limitation, a non-parametric bootstrap framework is implemented to simulate continuous test set variance and establish empirical 95% confidence intervals for the candidate models.

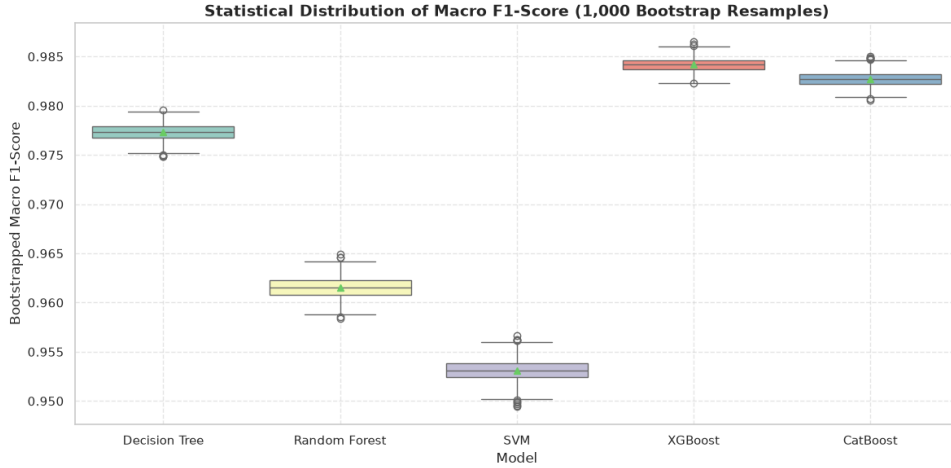


Figure 5.4: Empirical Macro F_1 -score probability density distributions generated via 1,000 independent non-parametric bootstrap resamples.

The resulting empirical metric distributions are plotted in Figure 5.4. The non-parametric resampling distribution demonstrates that the two tree-based gradient boosting architectures, XGBoost and CatBoost, exhibit highly compact and significantly overlapping density profiles. XGBoost secures the highest baseline performance with a Mean Macro F_1 -score of 0.9841 (95% CI: [0.9827, 0.9855]), while CatBoost closely mirrors this distribution with a Mean Macro F_1 -score of 0.9827 (95% CI: [0.9812, 0.9841]). The tight, overlapping confidence interval widths (~ 0.0028) confirm that both boosting frameworks achieve statistical equivalence on the cross-environmental testing days, validating their capacity to use sequential negative-gradient trees to isolate invariant propulsion metrics from non-stationary coastal environments.

The standalone Decision Tree model occupies the third-highest distribution tier, yielding a steady Mean Macro F_1 -score of 0.9773 (95% CI: [0.9756, 0.9788]), firmly outperforming the more complex Random Forest (Mean: 0.9615, 95% CI: [0.9594, 0.9635]) and soft-margin SVM architectures (Mean: 0.9531, 95% CI: [0.9508, 0.9553]).

While the Decision Tree slightly under performs the boosting configurations, its em-

pirical performance distribution provides a strong operational justification for hardware deployment. In automated passive monitoring frameworks, passive hydrophone systems are frequently deployed as isolated, battery-powered edge sensor nodes that face severe power consumption and memory constraints.

Advanced ensemble models like XGBoost or CatBoost require computing hundreds of sequential floating-point calculations per observation, while SVMs require intensive vector transformations. In contrast, a shallow, pruned Decision Tree can be directly compiled into simple, low-level execution logic (nested if-else conditional branches) that evaluates tabulated inputs with virtually zero computational overhead. Given that the standalone Decision Tree preserves an exceptionally high detection capability ($F_1 > 0.97$) with a highly stable confidence interval width, it represents an optimal operational architecture for real-time, low-power maritime surveillance applications at the embedded edge.

Chapter 6

Discussion & Conclusion

This chapter contextualises the empirical results of this study, addresses the underlying theoretical constraints of the current dataset, and outlines a structured roadmap for future research.

6.1 Proven Empirical Findings

The empirical results achieved across this investigation validate the utility of combining structured feature-engineering pipelines with classical machine learning architectures for uncooperative passive acoustic surveillance. By mapping raw multi-node hydrophone waveforms into distinct processing domains including LOFAR, DEMON, and noise-normalized broadband fields the classification algorithms successfully extracted discriminative mechanical and spectral attributes. The extraction and subsequent compression of the input space to 76 optimized acoustic predictors proved sufficient for isolating vessel signatures from non-stationary background noise fields without requiring computationally intensive deep learning models.

Methodologically, this research demonstrates that classical ensemble models can achieve high classification fidelity when protected by robust preprocessing and validation constraints. The tuned CatBoost and XGBoost configurations achieved global validation accuracies of 98.27% and 98.41%, respectively. More importantly, when evaluated via precision and recall metrics, the uniform integration of target validation confidence scores anchored the predictive behavior of the models. The confidence-weighted XGBoost framework achieved a Class 1 Precision of 0.99 and a Class 0 Recall of 0.99. This shift proves that anchoring the optimization objective to physical certainty metrics allows the classifier to suppress false alarms driven by continuous biological noise and localized environmental changes.

Furthermore, the stability of these metrics across the evaluation partition validates the day-separated block validation design. Because the testing data was drawn from complete, independent 24-hour intervals excluded from the training phase, the models demonstrated genuine cross-environmental generalisation. The pipeline successfully maintained its detection fidelity on unseen days characterized by shifting wind-driven noise floors and diurnal biological cycles. This structural independence indicates that the underlying acoustic predictors capture invariant physical characteristics of ship-radiated noise rather than overfitting to localized, short-term temporal anomalies.

6.2 Remaining Uncertainties & Theoretical Limitations

Despite the high performance metrics recorded across the ensemble configurations, several remaining uncertainties and theoretical limitations prevent this framework from being deemed fully ready for operational deployment. The predictive models were trained and evaluated on acoustic data captured during a single, constrained survey window spanning from 20 February to 4 March 2020. This narrow temporal window limits the dataset’s ability to reflect long-term environmental variations. Shifting seasonal sound speed profiles (SSPs), driven by fluctuations in water temperature, thermocline depth, and seasonal salinity gradients within the Hauraki Gulf, remain unobserved. Consequently, the cross-seasonal generalisation capability of the classifiers remains unquantified and requires extensive longitudinal validation before deployment in active naval surveillance networks.

From a hardware and architectural perspective, the deployment topology presents structural tracking constraints. The three Soundtrap 300HF logging units operate as isolated, omni-directional single sensor nodes. Because cross-channel sensor phase synchronization was not maintained during the initial survey, the current framework cannot support spatial array processing or acoustic beamforming. The individual stations record scalar pressure variations over a combined field, rendering the system incapable of resolving a target’s continuous spatial azimuth or multi-source tracking bearings without cooperative tracking data.

A Note on Data-Handling Limitations: While data leakage was mitigated by discarding acoustic features that directly generated the target assignments, the baseline reliance on historical AIS telemetry introduces an unquantified margin of error. The spatial-acoustic labelling heuristics depend on interpolated GPS broadcasts, which inherit dynamic transmission intervals and spatial alignment lags near a vessel’s Closest Point of Approach (CPA). This lag introduces localized label noise at the temporal onset and termination boundaries of an encounter. Furthermore, the selection bias inherent to AIS [8] means smaller recreational watercraft or illicit uncooperative vessels are under-represented in the high-confidence training brackets, forcing the labelling pipeline to rely on acoustic proxies that remain sensitive to environmental background fluctuations.

6.3 Future Work Framework

To address the theoretical limits of the current pipeline and transition this baseline binary framework toward a comprehensive maritime surveillance architecture, subsequent research should prioritize the following technical objectives:

- **Advanced Feature Space Refinement:** Implement advanced feature selection workflows, such as permutation feature importance mapping and greedy forward or beam selection selection pipelines, to compress the remaining 76-dimensional feature matrix into a streamlined subset optimized for embedded real-time hardware.
- **Longitudinal and Seasonal Validation:** Expand the acoustic data collection campaigns across a complete annual cycle to capture full seasonal Sound Speed Profile variations, ensuring the models are regularised against shifting oceanographic and thermocline boundaries.
- **Migration to Multi-Class Classification Setup:** Expand the task boundaries from a binary signal-versus-noise baseline detector into a multi-class configuration designed to isolate, identify, and categorize distinct vessel dimensions, propulsion types, or structural hull signatures.
- **Phase-Synchronized Array Integration:** Transition from isolated omni-directional logging nodes to phase-synchronized Horizontal or Vertical Line Arrays (VLAs) to enable adaptive spatial beamforming, allowing the system to isolate spatial sectors and calculate true target tracking bearings.
- **Cross-Sensor Data Fusion Networks:** Integrate the uncooperative passive acoustic classification pipeline with independent non-acoustic surveillance sensors, including shore-based marine radar and synthetic aperture satellite imagery (SAR), to dynamically verify dark vessel detections in real time.
- **Algorithmic Label Denoising and Confident Learning:** Implement label-noise mitigation frameworks, such as confident learning [11], to identify and correct faulty training labels caused by circular logic and timing lags near a vessel’s closest point of approach. Because automated labeling rules inevitably introduce errors at transit boundaries, using an algorithm to audit the dataset and remove or re-weight mislabeled samples prevents the models from memorizing flaws in the rules themselves. This ensures the optimization process anchors strictly to the genuine physical characteristics of ship-radiated noise rather than rule-based artifacts.

Summary Reflection

In summary, this project successfully establishes a mathematically sound, leakage-free baseline methodology for automated underwater vessel detection using classical machine learning. While longitudinal validation is required to ensure robust performance across seasonal environmental shifts, the confidence-weighted architecture demonstrates a reliable capacity to counter uncooperative tracking vulnerabilities, providing a reproducible blueprint for scalable acoustic monitoring across New Zealand’s strategic coastal choke points and territorial waters.

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Appendix A

Mathematical Foundations of Feature Extraction Blocks

This appendix provides the formal mathematical derivations, windowing functions, and structural signal processing transformations utilized within the automated DST data tabulation pipeline engineered by Sohan Ghodla. These formulations underpin the acoustic features analyzed by the machine learning classifiers.

A.1 LOFAR Spectral Transformations

To map the raw time-domain signal $x(n)$ into the low-frequency spectral space, the processing pipeline executes a discrete Short-Time Fourier Transform (STFT):

$$X(m, k) = \sum_{n=0}^{N-1} x(n + mH)w(n)e^{-j\frac{2\pi}{N}kn} \quad (\text{A.1})$$

where $w(n)$ represents a non-linear Hann window function of length N , H denotes the hop size or temporal step interval, and k represents the discrete frequency bin index. To ensure statistical stability against ambient fluctuations, the pipeline computes a median-averaged Welch Power Spectral Density (PSD) estimator over a localized bandpass range spanning 15 Hz to 2 kHz.

A.2 DEMON Envelope Demodulation Mechanics

To extract low-frequency amplitude modulation properties from high-frequency broadband cavitation noise carriers, the pipeline isolates the optimal carrier band using a Spectral Kurtosis ($K(f)$) maximization algorithm, defined via the fourth-order standardized moment:

$$K(f) = \frac{\langle |X(t, f)|^4 \rangle}{\langle |X(t, f)|^2 \rangle^2} - 2 \quad (\text{A.2})$$

where $\langle \cdot \rangle$ represents the temporal expectation operator across the localized time-frequency blocks. Following bandpass filtering through the high-kurtosis carrier allocation (1 kHz to 7.5 kHz), the signal is transformed via a continuous Hilbert transform to construct the complex analytic signal $z(t)$:

$$z(t) = x(t) + j\mathcal{H}\{x(t)\} = x(t) + j \left(\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \right) \quad (\text{A.3})$$

where $\mathcal{P}$ represents the Cauchy principal value. The low-frequency modulation envelope $E(t)$ is extracted as the absolute magnitude of this complex representation:

$$E(t) = |z(t)| = \sqrt{x^2(t) + [\mathcal{H}\{x(t)\}]^2} \quad (\text{A.4})$$

A secondary discrete Fourier transform mapped over $E(t)$ resolves cyclic shaft and blade frequencies within a specialized 1.0 Hz to 70 Hz analysis space.

A.3 Broadband Energy Integration Profiles

The absolute integrated acoustic energy (E_B) within defined upper and lower frequency sub-band boundaries $[f_1, f_2]$ is computed directly from the un-normalized Welch PSD estimate $S_x(f)$ via:

$$E_B = \int_{f_1}^{f_2} S_x(f) df \quad (\text{A.5})$$

Pairwise band-energy log-differences calculate the continuous spectral tilt indicator (α_{tilt}), while a first-order linear regression modeled across log-PSD versus log-frequency coordinates resolves the aggregate broadband energy decay slope.

A.4 Noise-Normalization and Spectral Whitening

To eliminate localized channel bias and environmental fluctuations from the data matrix, the active chunk power distribution $S_{\text{current}}(f)$ is whitened bin-by-bin using an environmental noise floor baseline $\tilde{S}_{\text{ambient}}(f)$:

$$S_{\text{whitened}}(f) = \frac{S_{\text{current}}(f)}{\tilde{S}_{\text{ambient}}(f)} \quad (\text{A.6})$$

where $\tilde{S}_{\text{ambient}}(f)$ is calculated via the 15th percentile of a 10-minute historical rolling spectral buffer $\mathcal{B}$.

A.5 Frame-Level Spectral Information Metrics

The statistical distribution of the raw acoustic energy spectrum is tracked using the Wiener Flatness coefficient (WF), which is evaluated as the ratio of the geometric mean to the arithmetic mean across K discrete frequency bins:

$$WF = \frac{\exp\left(\frac{1}{K} \sum_{k=1}^K \ln S(k)\right)}{\frac{1}{K} \sum_{k=1}^K S(k)} \quad (\text{A.7})$$

Concurrently, the spectral chaos distribution is tracked using the normalized Shannon spectral entropy (H_{spec}):

$$H_{\text{spec}} = -\frac{1}{\log_2 K} \sum_{k=1}^K p(k) \log_2 p(k) \quad (\text{A.8})$$

where $p(k) = S(k)/\sum S(i)$ represents the normalized spectral probability density of the k -th bin.

A.6 Mel-Frequency Cepstral Coefficients (MFCCs)

To model complex acoustic timbres, the Mel-warped log-energy outputs $Y(m)$ are projected onto a compact set of orthogonal vectors via a Discrete Cosine Transform (DCT-II):

$$c(n) = \sum_{m=1}^M \log_{10}[Y(m)] \cos\left(\frac{\pi n}{M} \left(m - \frac{1}{2}\right)\right) \quad (\text{A.9})$$

where M represents the total number of Mel filterbanks, and $c(n)$ is the resulting cepstral coefficient.

A.7 Temporal Waveform Dynamics

The temporal Zero-Crossing Rate (ZCR) is extracted directly from the raw time-series amplitude waveform $x(n)$ to preserve immediate high-frequency signal dynamics:

$$ZCR = \frac{1}{2(N-1)} \sum_{n=1}^{N-1} |\text{sgn}[x(n)] - \text{sgn}[x(n-1)]| \quad (\text{A.10})$$

where $\text{sgn}[\cdot]$ represents the standard signum step function.

Appendix B

Feature Dictionary and Schema Specifications

This appendix contains the exhaustive schema definitions and structural properties of the raw tabulated multi-node dataset. The raw dataset is structured as a `pandas` DataFrame utilizing a multi-level column index (`MultiIndex`) to maintain absolute separation between metadata, target labels, and processed model inputs, as summarized in Table B.1.

B.1 Dataframe Column MultiIndex Typology

Table B.1 establishes the top-level structural division of the data matrix columns before detailing individual variables.

Table B.1: DataFrame MultiIndex Column Hierarchy

Level Descriptor	Column Groups / Scopes	Operational Description
meta	chunk_start_utc, chunk_end_utc, phone	Temporal boundaries and physical hardware sensor identifiers.
lbl	label, label_confidence, ...	Target generation indicators, confidence weights, and telemetry.
feat	<i>(All acoustic processing blocks)</i>	Multi-dimensional input feature vectors evaluated by downstream models.

B.2 Labeling and Ground-Truth Matrix Schema (1b1)

Table B.2 itemizes the tracking vectors and spatial-acoustic metrics utilized by the rule-based labelling heuristics to formulate the target labels.

Table B.2: Metadata and Labelling Column Specifications (1b1)

Column Identifier	Physical and Operational Description
label	Binary target classification indicator: 0 = pure ambient background noise, 1 = active vessel encounter.
label_confidence	Continuous confidence weight tracking assignment certainty, bounded as $w_i \in (0, 1]$.
label_reason	Human-readable alphanumeric string detailing the specific spatial or acoustic rule-heuristic that triggered the label assignment.
n_vessels_in_box	Integer count tracking total concurrent AIS-active targets verified within the spatial operational search radius ($R_{\max}$).
M_AIS	Binary spatial telemetry mask indicating if an AIS-active target is physically present within the hydrophone basin (1 = True).
acoustic_proxy_norm	Normalized acoustic broadband proxy signal-to-noise ratio mapped continuously between $[0, 1]$.
closest_vessel_range_km	Continuous scalar tracking the geometric distance to the nearest AIS-broadcasting vessel position in kilometers (km).

B.3 Acoustic Feature Space Specifications (feat)

B.3.1 LOFAR Feature Blocks (Tonal and Narrowband Descriptors)

Table B.3 outlines the features extracted from the median-averaged Welch Power Spectral Density (PSD) to characterize discrete narrow-band line spectra.

Table B.3: LOFAR Tonal and Narrowband Feature Definitions

Feature Variable Name	Acoustic Signal Characteristic Description
lofar_peak_count	Discrete count of resolved narrow-band tonal peaks detected across the whitened spectrum.
lofar_broadband_snr_db	Integrated LOFAR-band signal-to-noise ratio evaluated against the historical noise baseline (dB).
lofar_max_snr_db	Maximum single frequency bin signal-to-noise ratio observed within the spectrum (dB).
lofar_dom_hz	Absolute frequency location of the most prominent, dominant narrow-band tonal peak (Hz).
lofar_dom_snr_db	Individual signal-to-noise ratio calculated for the dominant spectral peak (dB).
lofar_est_f0_hz	Estimated mechanical fundamental frequency (f_0) resolved via harmonic comb match filtering.
lofar_prom_sum	Summed mathematical prominence values across all resolved narrow-band spectral peaks.
lofar_peak_spacing_mean	Mean frequency distance separating sorted consecutive peak components (Hz).
lofar_peak_spacing_cv	Coefficient of variation of peak spacing; low variations reflect stable harmonic patterns.
lofar_hnr_db	Harmonic-to-Noise Ratio measuring active peak energy against the underlying continuum floor (dB).
lofar_snr_centroid	Spectral centroid tracking the center-of-mass distribution of the whitened spectrum.
lofar_snr_bandwidth	Spectral bandwidth measuring the standard deviation distribution around the spectral centroid.
lofar_snr_roll50	50% spectral roll-off frequency dividing the low and high energy distribution boundaries.
lofar_snr_roll95	95% spectral roll-off frequency isolating the upper boundary of structural spectral energy.
lofar_snr_flatness	Wiener flatness descriptor tracking the ratio of the geometric mean to the arithmetic mean.
lofar_snr_entropy	Normalized Shannon entropy tracking information uncertainty across the whitened power distribution.
lofar_snr_peakfreq	Exact frequency location of the maximum absolute power bin in the whitened PSD matrix.
lofar_peak_{1..6}_hz	Ordered frequency coordinate for the k -th strongest narrow-band machinery tonal.
lofar_peak_{1..6}_snr_db	Individual signal-to-noise ratio calculated for the k -th strongest machinery tonal (dB).

B.3.2 DEMON Feature Blocks (Propeller Blade-Rate Modulations)

Table B.4 details the features extracted via Hilbert envelope demodulation to track cyclic propulsion mechanics and propeller shaft dynamics.

B.3.3 Broadband Energy and Noise-Normalized Shape Blocks

Table [B.5](#) itemizes the parameters tracking absolute energy levels alongside whitened spectral shape variations across targeted sub-bands.

B.3.4 Frame-Level Spectral Descriptors, Cepstral Texture, and Waveform Dynamics

Table [B.6](#) compiles descriptors tracking short-term time-frame stability, Mel-scale audio timbres, and immediate raw time-series amplitude profiles.

Table B.4: DEMON Amplitude Modulation Feature Definitions

Feature Variable Name	Acoustic Modulation Characteristic Description
cav_lo	Lower frequency boundary of the adaptive high-frequency cavitation carrier band (Hz).
cav_hi	Upper frequency boundary of the adaptive high-frequency cavitation carrier band (Hz).
env_*	Baseline temporal statistics (<i>mean, std, min, p10, p50, p90, max</i>) of the raw envelope stream.
env_skew	Third-order statistical skewness measuring asymmetry across the raw temporal envelope.
env_kurt	Fourth-order statistical kurtosis tracking transient impulsiveness in the raw temporal envelope.
env_rms	Root-Mean-Square energy calculation evaluating the total power of the temporal envelope.
env_crest	Crest factor ratio of peak envelope amplitude relative to the localized envelope RMS level.
env_rel_*	Comparative metrics (<i>mean, std, etc.</i>) scaled by the chunk mean to yield gain-independent signatures.
demon_mod_index	Global Temporal Modulation Index, defined mathematically via the ratio $\sigma_{\text{env}}/\mu_{\text{env}}$.
demon_broadband_snr_db	Integrated modulation band signal-to-noise ratio computed over the 1–70 Hz modulation spectrum.
demon_max_snr_db	Maximum single frequency modulation bin signal-to-noise ratio observed (dB).
demon_snr_centroid	Spectral centroid evaluating the center-of-mass across the whitened DEMON spectrum.
demon_snr_bandwidth	Standard deviation distribution width spanning the whitened modulation spectrum centroid.
demon_snr_roll50	50% energy roll-off frequency coordinate across the whitened modulation matrix.
demon_snr_roll95	95% energy roll-off frequency coordinate across the whitened modulation matrix.
demon_snr_flatness	Wiener flatness tracking tonal modulation structure versus flat background noise modulations.
demon_snr_entropy	Normalized Shannon entropy calculating spectral randomness across the modulation spectrum.
demon_snr_peakfreq	Modulation frequency location tracking the maximum absolute whitened DEMON energy bin.
demon_peak_count	Total discrete count of operational modulation peaks resolved within the blade-rate window.
demon_dom_hz	Resolved frequency of the dominant modulation peak, tracking primary blade-rate signatures (Hz).
demon_dom_snr_db	Individual signal-to-noise ratio calculated for the dominant modulation frequency peak (dB).
demon_est_f0_hz	Estimated propeller shaft fundamental rotational frequency (f_0) via harmonic comb optimization.
demon_prom_sum	Summed mathematical prominence values across all resolved modulation frequency peaks.
demon_acf_peak_lag_s	Temporal lag location tracking the strongest non-zero peak in the autocorrelation matrix (seconds).
demon_acf_peak_hz	Equivalent cyclical frequency corresponding directly to the primary autocorrelation peak lag (Hz).
demon_acf_peak_height	Absolute mathematical height value of the primary autocorrelation peak, bounded between $[0, 1]$.
demon_acf_regularity	Structural consistency ratio computed via the height difference between the secondary and primary peaks.

Table B.5: Absolute and Noise-Normalized Broadband Feature Definitions

Feature Variable Name	Acoustic Energy and Shape Characteristic Description
Absolute Energy Descriptors (Un-normalized Welch PSD)	
bb_energy_total_db	Integrated global mean PSD energy computed across the full operational bandwidth (dB).
bb_energy_flow_db	Localized hydrophone flow-noise sub-band energy spanning 10–40 Hz (dB).
bb_energy_low_db	Low-frequency machinery and hull propulsion sub-band energy spanning 40–250 Hz (dB).
bb_energy_mid_db	Mid-frequency auxiliary system and cavitation sub-band energy spanning 250–1500 Hz (dB).
bb_energy_high_db	High-frequency environmental and biological sub-band energy spanning 1500–6000 Hz (dB).
bb_balance_*	Pairwise differences between sub-band absolute energies, tracking raw spectral tilt profiles.
bb_rolloff_{25,50..95}	Frequencies below which 25%, 50%, 75%, 85%, and 95% of total absolute energy resides.
bb_spectral_slope	Linear regression slope calculated across the log-PSD versus log-frequency coordinate space.
Noise-Normalized Shape Descriptors (Whitened Spectrum)	
snr_broadband_db	Total integrated signal-to-noise ratio resolved over the fully whitened spectrum (dB).
snr_band_flow_db	Whitened flow-noise sub-band signal-to-noise ratio spanning 10–40 Hz (dB).
snr_band_low_db	Whitened low-frequency machinery signature signal-to-noise ratio spanning 40–250 Hz (dB).
snr_band_mid_db	Whitened mid-frequency auxiliary machinery signal-to-noise ratio spanning 250–1500 Hz (dB).
snr_band_high_db	Whitened high-frequency environmental signature signal-to-noise ratio spanning 1500–6000 Hz (dB).
snr_band_very_high_db	Whitened extreme high-frequency acoustic tracking signal-to-noise ratio above 6000 Hz (dB).
snr_balance_*	Pairwise difference calculations between whitened sub-bands, capturing normalized spectral shifts.
snr_rolloff_{25..95}	Energy integration boundaries evaluated continuously across the whitened signal spectrum.
snr_spectral_slope	Calculated linear decay slope of the whitened signal vector versus the log-frequency axis.

Table B.6: Spectral Metrics, MFCCs, and Temporal Waveform Feature Definitions

Feature Variable Name	Micro-Acoustic and Waveform Characteristic Description
Spectral Metrics (Computed Frame-by-Frame over 20–2000 Hz)	
spectral_flatness_*	Wiener entropy frame summaries (<i>mean, std, min, max, percentiles</i>); 0 = tonal, 1 = noise.
spectral_entropy_*	Normalized Shannon entropy frame distribution summaries, tracking spectral texture chaos.
MFCC Descriptors (13 Orthogonal Cepstral Coefficients via 12 kHz Decimation)	
mfcc_{1..13}_mean	Temporal mean of the k -th cepstral coefficient across the processing chunk.
mfcc_{1..13}_std	Standard deviation of the k -th cepstral coefficient, capturing structural timbral shifting.
Temporal Features (Extracted Direct From Raw Amplitude Waveforms)	
zcr_*	Statistical summary parameters tracking the time-domain Zero-Crossing Rate.
rms_*	Frame-level Root-Mean-Square tracking summaries, resolving short-term power envelope shifts.
rms_overall	Continuous global Root-Mean-Square calculation tracking the absolute power of the full acoustic chunk.
crest_factor	Waveform peak-to-RMS ratio measuring transient sharpness against continuous background noise.
tmi_cav	Temporal Modulation Index computed exclusively within the high-SK cavitation carrier band.